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Multi-Currency Portfolio Optimization with General Risk Measures

Master Thesis

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Acknowledgments

Abstract

Multi-currency portfolio construction requires managing asset and currency exposures together, yet it remains unclear whether these decisions are best made simultaneously or sequentially as the investable universe grows. This thesis builds on the regularized multi-currency Expected Shortfall framework of Lucescu and Ulrych (2026), in which portfolios are formed by minimizing a risk objective subject to L^1 and L^2 penalties, with currency exposures either optimized jointly with assets or handled through a separate overlay. First we replicate the framework's core strategies, then extend it along two dimensions, the size of the investable universe, and the choice of objective function, implementing both Expected Shortfall and mean-variance formulations of the joint and overlay optimizers.

Across both objectives, the joint optimizer's out-of-sample performance deteriorates relative to the separate overlay as the universe expands. This motivates the central question of the thesis, is this deterioration a structural property of joint optimization, or is it modulated by the choice of objective function? To answer it, we develop a general regularized optimization framework that admits arbitrary objective functions within a single, common formulation. We validate the framework by showing that it exactly reproduces the existing Expected Shortfall and mean-variance optimizers, and then use it to measure how the rate of the joint optimizer's breakdown varies as dimensionality increases.

Sentence on results.

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Chapter 1

Introduction

1.1 Motivation: International Diversification, Currency Risk, and Tail Risk

1.2 Contributions

1.3 Thesis Structure

Chapter 2

Literature Review

Cite the following: International allocation and currency risk management, Currency overlay vs. joint optimization, Downside risk measures and Expected Shortfall, Instability of ES optimization and the role of regularization, Generalized objectives and decision-focused / end-to-end learning.

In this chapter, we review the relevant literature on multi-currency portfolio optimization. We discuss the contributions and methodologies in the field, highlighting how our work builds upon and extends existing research.

The core of our analysis builds on two recent studies that apply regularization techniques to international asset allocation. Burkhardt and Ulrych (2023) introduce a joint optimization approach within a mean-variance framework. By using L1 and L2 regularization, they show that jointly optimizing assets and currencies leads to better out-of-sample performance than the standard industry practice of using a separate currency overlay.

Extending this framework, Lucescu and Ulrych (2026) consider an Expected Shortfall optimization framework to better manage downside tail risk. To fix the numerical instability common in Expected Shortfall minimization, they apply similar regularization methods. Their findings reveal a dimensional trade-off. Joint optimization works best in small asset universes, but the separate currency overlay approach becomes more effective for larger universes due to its built-in dimensionality reduction.

Reproducing and comparing these two baseline frameworks serves as the starting point for the extensions proposed in this thesis.

Chapter 3

Optimization Framework

This chapter develops the optimization framework used throughout the thesis, building up from a model of multi-currency portfolio returns to the risk objectives, the regularization that stabilizes them, and the two formulations through which asset and currency decisions are combined. The derivations follow the methodology presented in [4] and [3], use simple returns, and assume that rebalancing occurs at the end of each investment period. We begin in Section 3.1 with a model-free representation of one-period portfolio returns for an investor who holds assets denominated in multiple currencies and can take currency forward positions for hedging, decomposing the unhedged return of a foreign asset into its local, currency, and cross-currency components, introducing the currency-mapping matrix that links assets to their implicit currency exposures, and expressing the hedged portfolio return in terms of net currency exposures.

With the return model in place, we formulate the optimization problem. Section 3.2 states the classical mean-variance criterion, and Section 3.3 introduces Expected Shortfall as a coherent downside-risk measure, together with the Rockafellar–Uryasev representation that casts its minimization as a tractable linear program. Both are presented as instances of a common mean–risk objective, so that the framework treats the risk measure as a swappable component rather than a fixed choice. Section **??** then introduces the L_1 and L_2 penalties and the leverage constraint used to stabilize the optimization and improve out-of-sample performance.

Finally, Sections **??** and **??** assemble these elements into the two approaches to multi-currency allocation studied throughout the thesis: the separate (currency overlay) formulation, in which asset and currency decisions are made sequentially, and the joint formulation, in which they are optimized simultaneously. The detailed derivations of the resulting quadratic programs are found in Appendix A and Appendix B.

3.1 Model-Free Multi-Currency Portfolio Returns

3.1.1 Unhedged Portfolio Returns

We consider an international investor who allocates their wealth across N assets denominated in up to $M+1$ currencies. We fix a discrete time index $t \in \{0, 1, 2, \dots\}$ and one investment period from t to $t{+}1$. Let the investor's *domestic currency* be indexed by 0 and let the remaining M foreign currencies be indexed by $\mathcal{C} \coloneqq \{1, 2, \dots, M\}$.

Let $P_{i,t} > 0$ be the price of asset i at time t in its local currency $c_i \in \mathcal{C} \cup \{0\}$. Define the

corresponding *simple local-currency asset return* over $(t, t{+}1]$ as

$$R_{i,t+1} \coloneqq \frac{P_{i,t+1} - P_{i,t}}{P_{i,t}}.$$

Thus $P_{i,t+1} = P_{i,t}(1 + R_{i,t+1})$.

For each foreign currency $c \in \mathcal{C}$, let $S_{c,t} > 0$ denote the exchange rate at time t, quoted as units of domestic currency per one unit of foreign currency c. The associated *simple currency return* over $(t, t{+}1]$ is

$$e_{c,t+1} \coloneqq \frac{S_{c,t+1} - S_{c,t}}{S_{c,t}}.$$

Equivalently, $S_{c,t+1} = S_{c,t}(1+e_{c,t+1})$. For the domestic currency, we set $S_{0,t} \equiv 1$ and $e_{0,t+1} \equiv 0$.

The domestic-currency value of holding one unit of asset i at time t is $P_{i,t}\, S_{c_i,t}$. Therefore, the *unhedged domestic-currency simple return* of asset i is

$$\begin{aligned} R^u_{i,t+1} &= \frac{P_{i,t+1}S_{c_i,t+1} - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= \frac{P_{i,t}(1 + R_{i,t+1})\, S_{c_i,t}(1 + e_{c_i,t+1}) - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= R_{i,t+1} + e_{c_i,t+1} + R_{i,t+1}\, e_{c_i,t+1}. \end{aligned} \tag{3.1}$$

Let $\boldsymbol{x}_t \in \mathbb{R}^N$ be the vector of asset portfolio weights chosen at time t and held until $t{+}1$, where $x_{i,t}$ is the fraction of total portfolio value invested in asset i at time t. We impose the standard budget constraint $\mathbf{1}^\mathsf{T}\boldsymbol{x}_t = 1$.

Assuming no rebalancing between times t and $t{+}1$, the domestic-currency portfolio's unhedged return is given by

$$R^u_{\mathcal{P},t+1} = \sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}.$$

3.1.2 Hedged Portfolio Returns

To manage currency risk, the investor can take positions in one-period currency forward contracts. For each foreign currency $c \in \mathcal{C}$, let $F_{c,t}$ be the forward rate quoted at time t for settlement at $t{+}1$, in units of domestic currency per unit of foreign currency c. We define the *forward premium* by

$$f_{c,t} \coloneqq \frac{F_{c,t} - S_{c,t}}{S_{c,t}}.$$

The investor hedges by selling foreign currency forward. Let $\phi_{c,t}$ denote the relative notional of the *short* forward position in currency c at time t, expressed as a fraction of total portfolio value, with the convention that $\phi_{c,t} > 0$ corresponds to hedging. Per unit of notional, the short position pays $f_{c,t} - e_{c,t+1}$ in domestic currency over the period. Adding these forward payoffs to the unhedged asset returns, the *hedged portfolio return* is

$$R^h_{\mathcal{P},t+1} = \underbrace{\sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}}_{R^u_{\mathcal{P},t+1}} + \sum_{c=1}^{M} \phi_{c,t}\,(f_{c,t} - e_{c,t+1}). \tag{3.2}$$

3.1.3 Matrix Representation and Decomposition

To express the returns in a compact form suitable for optimization, we define the currency incidence matrix $\boldsymbol{C} \in \{0,1\}^{N\times M}$ by

$$C_{i,c} \coloneqq \begin{cases} 1, & \text{if asset } i \text{ is denominated in foreign currency } c, \\ 0, & \text{otherwise.} \end{cases}$$

Each row of $\boldsymbol{C}$ has at most one nonzero entry, and the rows corresponding to domestic assets are identically zero. This allows us to define the portfolio's *implicit foreign-currency exposure* as

$$\boldsymbol{w}_t \coloneqq \boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t \in \mathbb{R}^M, \qquad w_{c,t} = \sum_{i:\, c_i=c} x_{i,t},$$

such that $w_{c,t}$ is the total portfolio weight invested in assets denominated in currency c. The implicit domestic exposure follows residually as $w_{0,t} = 1 - \mathbf{1}_M^\mathsf{T}\boldsymbol{w}_t$, since the asset weights sum to one.

We express returns in vector form $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$, $\boldsymbol{e}_{t+1} \in \mathbb{R}^M$, and the forward premiums as $\boldsymbol{f}_t \in \mathbb{R}^M$. The product $\boldsymbol{C}\boldsymbol{e}_{t+1} \in \mathbb{R}^N$ aligns currency returns to assets, with ith component $e_{c_i,t+1}$ for foreign assets and 0 for domestic ones. Using (3.1), the vector of *unhedged domestic returns* across assets can be written as

$$\boldsymbol{R}^u_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}),$$

where $\odot$ denotes the Hadamard product.

Expressing the forward positions in vector form as $\boldsymbol{\phi}_t \in \mathbb{R}^M$, the hedged portfolio return (3.2) can be written as

$$R^h_{\mathcal{P},t+1} = \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + \boldsymbol{\phi}_t^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}). \tag{3.3}$$

Setting $\boldsymbol{\phi}_t = \boldsymbol{w}_t = \boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t$ neutralizes the implicit currency exposure. Substituting this together with the unhedged return vector into (3.3), the *fully hedged portfolio return* is

$$\begin{aligned} R^{fh}_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + (\boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t)^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1})\Big] + \boldsymbol{x}_t^\mathsf{T}\boldsymbol{C}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}) + \boldsymbol{C}\boldsymbol{f}_t - \boldsymbol{C}\boldsymbol{e}_{t+1}\Big] \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1})\Big] \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^{fh}_{t+1}. \end{aligned}$$

The first-order currency terms $\pm\boldsymbol{x}_t^\mathsf{T}\boldsymbol{C}\boldsymbol{e}_{t+1}$ cancel, leaving the *fully hedged asset return vector*

$$\boldsymbol{R}^{fh}_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}).$$

More generally, define the *net currency exposure* $\boldsymbol{\psi}_t \coloneqq \boldsymbol{w}_t - \boldsymbol{\phi}_t$. Imposing $\sum_{c=0}^M \phi_{c,t} = 1$ gives $\sum_{c=0}^M \psi_{c,t} = 0$, so the currency overlay is a zero-investment portfolio. Substituting $\boldsymbol{\phi}_t = \boldsymbol{w}_t - \boldsymbol{\psi}_t$ into (3.3) expresses the hedged return in terms of net exposures,

$$\begin{aligned} R^h_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + (\boldsymbol{w}_t - \boldsymbol{\psi}_t)^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + \boldsymbol{w}_t^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) + \boldsymbol{\psi}_t^\mathsf{T}(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^{fh}_{t+1} + \boldsymbol{\psi}_t^\mathsf{T}(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t), \end{aligned} \tag{3.4}$$

which we adopt throughout the following portfolio optimization formulations, as it makes the asset weights $\boldsymbol{x}_t$ and the net currency exposures $\boldsymbol{\psi}_t$ the explicit variables of the optimization.

3.2 Mean-Variance Optimization Formulation

3.2.1 The Mean-Variance Criterion

The mean-variance criterion of [2] measures portfolio risk by the variance of the portfolio return and selects weights that balance this risk against expected return. Using the hedged portfolio return $R^h_{\mathcal{P},t+1}$ defined in (3.4) and a relative risk-aversion parameter $\gamma > 0$, the investor solves

$$\min \ \frac{\gamma}{2}\operatorname{Var}\Big(R^h_{\mathcal{P},t+1}\Big) - \mathbb{E}\Big[R^h_{\mathcal{P},t+1}\Big], \qquad \text{s.t.} \quad \mathbf{1}_N^\top \boldsymbol{x}_t = 1. \tag{3.5}$$

The parameter γ governs the trade-off between variance and expected return, larger values place more weight on risk reduction, and in the limit $\gamma \to \infty$ the criterion reduces to pure variance minimization.

The hedged portfolio return (3.4) is linear in the asset weights $\boldsymbol{x}_t$ and the net currency exposures $\boldsymbol{\psi}_t$. Consequently $\mathbb{E}[R^h_{\mathcal{P},t+1}]$ is linear and $\operatorname{Var}(R^h_{\mathcal{P},t+1})$ is a quadratic form in these decision variables, so that, once the expected-return vector $\boldsymbol{\mu}$ and the covariance matrix $\boldsymbol{\Sigma}$ are estimated, (3.5) is a convex quadratic program. Whether the asset and currency weights are chosen sequentially or in a single step determines the precise decision vector and the accompanying constraints; these two formulations are developed in Sections 3.2.3 and 3.2.4.

While analytically convenient, variance is not a coherent risk measure in the sense of [1], who characterize a coherent measure by monotonicity, subadditivity, positive homogeneity, and translation invariance. Variance violates monotonicity: a portfolio whose return dominates another in every state of the world need not have lower variance, so variance can rank a uniformly preferable position as the riskier one. Equally relevant for international portfolios exposed to regime shifts and market stress, variance penalizes upside and downside deviations symmetrically and therefore does not isolate the large losses that most concern investors. These shortcomings motivate the coherent, downside-focused alternative developed in Section 3.3, where Expected Shortfall replaces variance as the risk measure while the surrounding optimization structure is left unchanged.

3.2.2 Regularized Mean-Variance Problem

3.2.3 Currency Overlay Optimization

3.2.4 Joint Optimization

3.3 Expected Shortfall Optimization Formulation

3.3.1 Expected Shortfall and the Rockafellar–Uryasev Representation

3.3.2 Regularized Expected Shortfall Problem

3.3.3 Currency Overlay Optimization

3.3.4 Joint Optimization

Chapter 4

Data and Implementation

4.1 Data and Summary Statistics

Our analysis relies on a comprehensive dataset of international equities, foreign exchange spot rates, and forward exchange rates sourced from Refinitiv Datastream. The dataset covers a period from June 1990 to June 2023, capturing various market regimes, including the dot-com bubble burst, the 2008 Global Financial Crisis, the European debt crisis, and the COVID-19 pandemic.

We construct our investment universe from the 14 largest developed equity markets: Australia, Canada, France, Germany, Hong Kong, Italy, Japan, the Netherlands, Singapore, Spain, Sweden, Switzerland, the United Kingdom, and the United States. To construct the local equity portfolios, we select the largest stocks in each market based on their market capitalization at the beginning of each year. We consider two distinct cross-sections of stocks to analyze the impact of the investment universe size on portfolio performance, a small universe consisting of the top 10 stocks per market (140 stocks in total) and a large universe consisting of the top 50 stocks per market (700 stocks in total).

For each selected stock, we retrieve daily total return indices, which account for dividend reinvestments, and compute the corresponding local currency returns. The US dollar serves as the base currency for our analysis, we calculate the fully hedged returns for each stock following the model-free return representation in chapter **??**.

4.2 Modular Backtesting Framework

This chapter describes the implementation of the walk-forward backtesting framework used throughout the thesis. The code is organised around a small number of clearly separated components: a data handler that prepares the investment universe and return series, a backtester that orchestrates the rolling evaluation, and a family of optimizers that implement the portfolio construction problem.[1] This modular structure keeps the experimental setup reproducible while allowing different objective functions and portfolio architectures to share the same evaluation pipeline.

The figure highlights the main design principle of the implementation: the backtester is intentionally kept generic, while the portfolio logic lives in interchangeable optimizer classes. This makes it possible to compare mean-variance and expected shortfall formulations, asset-

[1]The full codebase for this project is publicly available: https://github.com/anejrozman/Multi-Currency-Portfolio-Optimization-with-General-Risk-Measures.

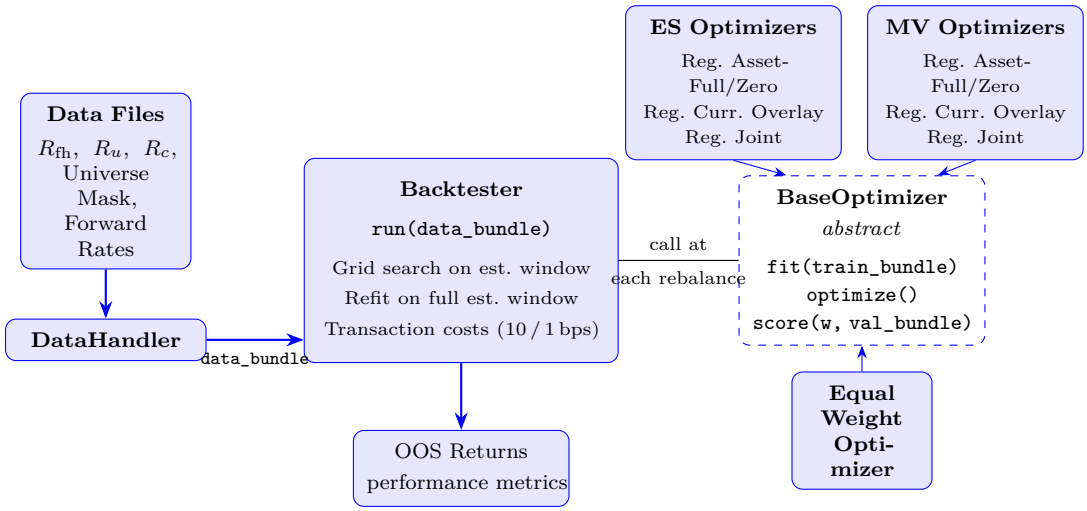


Figure 4.1: Schematic of the backtesting framework.

only and joint asset–currency strategies, and regularized and unregularized variants under the same walk-forward protocol.

4.2.1 Backtesting Workflow and Optimizer Structure

The framework is built around a strict separation of concerns. The `DataHandler` prepares the inputs for the backtest by loading precomputed return series, mapping each asset to its reporting currency, and building a daily universe mask from the investment universe file. The backtester receives these objects as a single `data_bundle` and does not need to know how they were generated.

The `Backtester` is responsible for the evaluation protocol. It determines rebalancing dates, constructs estimation and validation windows, performs hyperparameter selection when a grid is supplied, and records the resulting out-of-sample returns. All portfolio construction decisions are delegated to subclasses of `BaseOptimizer`, which all implement `fit(train_bundle)`, `optimize()`, and `score(weights, val_bundle)`.

The backtest follows a standard walk-forward procedure. At each rebalance date, the framework forms a rolling estimation window and a separate validation window, evaluates candidate hyperparameters, refits the best configuration on the full estimation sample, and then carries the portfolio into the next holding period. Rebalancing can be daily, weekly, monthly, quarterly, semi-annual, or annual. Before optimization, the code filters the investable universe to currently eligible assets and checks that each selected asset has a complete return history, which keeps missing-data handling outside the optimization problem. Window lengths are stored centrally in `backtest_config.py`; the default setup uses a ten-year estimation window and a one-year validation window.

4.2.2 Optimization, Costs, and Outputs

Two optimizer families are implemented: expected shortfall and mean-variance. Expected shortfall is formulated as a convex program with auxiliary variables, optional ℓ_1 and ℓ_2 regularization, and an ℓ_1-norm leverage cap. Mean-variance uses Bayes–Stein shrinkage for expected returns and a lightly regularized covariance matrix. Both families are available in asset-only and joint asset–currency variants. The asset-only strategies use either hedged returns (*assets-full*) or unhedged returns (*assets-zero*), while the joint strategies solve for asset

weights and currency hedge weights together. The equal-weight benchmark follows the same interface but returns a uniform allocation.

All substantive optimizers are solved with CVXPY. When available, Gurobi is used as the default solver with tight tolerances; otherwise, the code falls back to another compatible solver. Transaction costs are applied at each rebalance using separate assumptions for equities and FX, namely 10 basis points and 1 basis point. Costs are computed with a log-linear turnover approximation, and the backtester tracks drifted weights between rebalances so that turnover reflects actual portfolio evolution. The framework outputs daily out-of-sample returns and summary statistics including annualized volatility, maximum drawdown, Sharpe ratio, Sortino ratio, and an expected shortfall ratio.

4.3 Numerical Implementation of Optimization Problems

4.4 Evaluation Protocol and Performance Metrics

Chapter 5

Empirical Analysis and Baseline Results

In this chapter, we present an empirical analysis of the multi-currency portfolio optimization problem using historical data. We apply the model-free return representation derived in chapter **??** to compute realized returns for the portfolio strategies derived in section **??**. We evalutate the performance of the various strategies on two investment universes, a large investment universe consisting of 50 biggest stocks in the 14 largest developed markets and a small investment universe consisting of 10 biggest stocks in the same markets. We compare the performance of the optimized portfolios to a benchmark equally weighted portfolio.

5.1 Backtest Results of Expected Shortfall Optimization Approaches

We can see in table 5.1 that the performance of the expected shortfall optimization approaches varies across different universe sizes. In the small universe, the optimized portfolios generally outperform the equal weight benchmark in terms of risk-adjusted returns, as measured by the Sharpe Ratio and Sortino Ratio. However, as we increase the universe size to medium and large, the performance of the optimized portfolios relative to the equal weight benchmark deteriorates, with some strategies even underperforming the benchmark in terms of risk-adjusted returns. This suggests that the benefits of optimization may diminish as the investment universe expands, potentially due to increased estimation error and model uncertainty in larger universes.

5.2 Backtest Results of Mean-Variance Optimization Approaches

5.3 The Effects of Universe Size on Portfolio Performance

Panel A: Small Investment Universe (top 10 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel B: Medium Investment Universe (top 50 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel C: Large Investment Universe (top 100 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							

Table 5.1: Performance metrics for Expected Shortfall optimization approaches on increasing universe sizes

Table 5.1 reports out-of-sample performance metrics for the international portfolio optimization under expected shortfall, with portfolios rebalanced quarterly. The realized out-of-sample statistics shown are Expected Shortfall (ES), volatility (Vol), maximum drawdown (MD), the Sharpe Ratio (SR), the Sortino Ratio (SOR), the Expected Shortfall Ratio (ESR), and average asset and currency turnover, denoted Turnover^a and Turnover^c, respectively. Each panel corresponds to a different investment universe: Panel A covers the small universe, made up of the 10 largest companies by market capitalization in each economy; Panel B the medium universe, comprising the 50 largest companies per economy; and Panel C the large universe, consisting of the 100 largest companies per economy. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. Details on the optimization approaches and performance metrics are provided in section **??**.

Panel A: Small Investment Universe (top 10 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel B: Medium Investment Universe (top 50 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel C: Large Investment Universe (top 100 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							

Table 5.2: Performance metrics for mean-variance optimization approaches on increasing universe sizes

Table 5.2 reports out-of-sample performance metrics for the international portfolio optimization under mean-variance, with portfolios rebalanced quarterly. The realized out-of-sample statistics shown are Expected Shortfall (ES), volatility (Vol), maximum drawdown (MD), the Sharpe Ratio (SR), the Sortino Ratio (SOR), the Expected Shortfall Ratio (ESR), and average asset and currency turnover, denoted Turnover^a and Turnover^c, respectively. Each panel corresponds to a different investment universe: Panel A covers the small universe, made up of the 10 largest companies by market capitalization in each economy; Panel B the medium universe, comprising the 50 largest companies per economy; and Panel C the large universe, consisting of the 100 largest companies per economy. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. Details on the optimization approaches and performance metrics are provided in section **??**.

Chapter 6

Convex Risk Measures

Chapter 7

End To End Portfolio Optimization

Chapter 8

Conclusion

8.1 Findings

8.2 Limitations and Future Research

Appendices

Appendix A

Optimization Derivation: Expected Shortfall

We derive the equivalence between the direct minimization of Expected Shortfall and the auxiliary formulation proposed by Rockafellar and Uryasev (2000) that allows for formulating the optimization as a linear program.

We assume that the loss distribution is continuous, meaning the loss cumulative distribution function has no jumps and the probability density function $f(l)$ exists. Under these assumptions, Value-at-Risk (VaR) is uniquely defined.

Define the loss random variable at time $t+1$ given portfolio weights x_t as $L(x_t) = -x_t^T R_{t+1}$. Let $f(l)$ denote the probability density function of the loss $L(x_t)$. The VaR at confidence level β is defined as the β-quantile of the loss distribution

$$\mathrm{VaR}_\beta(x_t) = \min\{\alpha : \mathbb{P}(L(x_t) \leq \alpha) \geq \beta\}.$$

The Expected Shortfall is defined as the conditional expectation of losses exceeding VaR

$$\mathrm{ES}_\beta(x_t) = \mathbb{E}[L(x_t) \mid L(x_t) \geq \mathrm{VaR}_\beta(x_t)] = \frac{1}{1-\beta} \int_{\mathrm{VaR}_\beta(x_t)}^{\infty} l f(l)\, dl.$$

We now consider the auxiliary function F_β defined as

$$F_\beta(x_t, \alpha) = \alpha + \frac{1}{1-\beta} \mathbb{E}_t[(L(x_t) - \alpha)^+].$$

We can rewrite the expectation term using the integral form

$$\mathbb{E}_t[(L(x_t) - \alpha)^+] = \int_{-\infty}^{\infty} (l-\alpha)^+ f(l)\, dl = \int_{\alpha}^{\infty} (l-\alpha) f(l)\, dl.$$

Before minimizing $F_\beta(x_t, \alpha)$ with respect to α, we verify convexity. The second derivative of F_β with respect to α is

$$\begin{aligned} \frac{\partial F_\beta}{\partial \alpha} &= 1 - \frac{1}{1-\beta} \int_{\alpha}^{\infty} f(l)\, dl \\ \frac{\partial^2 F_\beta}{\partial \alpha^2} &= \frac{1}{1-\beta} f(\alpha). \end{aligned}$$

Since the probability density function $f(\alpha) \geq 0$ and $\beta < 1$, the second derivative is non-negative, $\frac{\partial^2 F_\beta}{\partial \alpha^2} \geq 0$. Thus, $F_\beta(x_t, \alpha)$ is convex in α, ensuring that the first-order condition is necessary and sufficient for a global minimum.

Differentiation of $F_\beta(x_t, \alpha)$ with respect to α yields the first-order condition for optimality. Using Leibniz's rule we have

$$\begin{aligned}
\frac{\partial F_\beta}{\partial \alpha} &= 1 + \frac{1}{1-\beta}\frac{\partial}{\partial \alpha}\int_\alpha^\infty (l-\alpha) f(l)\, dl \\
&= 1 + \frac{1}{1-\beta}\left[-f(\alpha)\cdot 0 + \int_\alpha^\infty (-1) f(l)\, dl\right] \\
&= 1 - \frac{1}{1-\beta}\int_\alpha^\infty f(l)\, dl \\
&= 1 - \frac{1}{1-\beta}\mathbb{P}(L(x_t) \geq \alpha).
\end{aligned}$$

Setting the derivative to zero to find the optimal α^*

$$\begin{aligned}
1 - \frac{1}{1-\beta}\mathbb{P}(L(x_t) \geq \alpha^*) &= 0 \\
\mathbb{P}(L(x_t) \geq \alpha^*) &= 1-\beta \\
\mathbb{P}(L(x_t) \leq \alpha^*) &= \beta.
\end{aligned}$$

This implies that the optimal α^* that minimizes the auxiliary function for a fixed x_t is exactly the Value-at-Risk at level β, so $\alpha^* = \mathrm{VaR}_\beta(x_t)$.

Substituting $\alpha^* = \mathrm{VaR}_\beta(x_t)$ back into $F_\beta(x_t, \alpha)$ we obtain

$$\begin{aligned}
F_\beta(x_t, \mathrm{VaR}_\beta(x_t)) &= \mathrm{VaR}_\beta(x_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(x_t)}^\infty (l - \mathrm{VaR}_\beta(x_t)) f(l)\, dl \\
&= \mathrm{VaR}_\beta(x_t) + \frac{1}{1-\beta}\left(\int_{\mathrm{VaR}_\beta(x_t)}^\infty l f(l)\, dl - \mathrm{VaR}_\beta(x_t)\int_{\mathrm{VaR}_\beta(x_t)}^\infty f(l)\, dl\right).
\end{aligned}$$

Recognizing that $\int_{\mathrm{VaR}_\beta(x_t)}^\infty f(l)\, dl = \mathbb{P}(L(x_t) \geq \mathrm{VaR}_\beta(x_t)) = 1-\beta$, we have

$$\begin{aligned}
F_\beta(x_t, \mathrm{VaR}_\beta(x_t)) &= \mathrm{VaR}_\beta(x_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(x_t)}^\infty l f(l)\, dl - \frac{1}{1-\beta}\mathrm{VaR}_\beta(x_t)(1-\beta) \\
&= \mathrm{VaR}_\beta(x_t) + \mathrm{ES}_\beta(x_t) - \mathrm{VaR}_\beta(x_t) \\
&= \mathrm{ES}_\beta(x_t).
\end{aligned}$$

Thus, minimizing $\mathrm{ES}_\beta(x_t)$ with respect to x_t is equivalent to minimizing $F_\beta(x_t, \alpha)$ jointly with respect to x_t and α

$$\min_{x_t} \mathrm{ES}_\beta(x_t) = \min_{x_t, \alpha}\left(\alpha + \frac{1}{1-\beta}\mathbb{E}_t[(-x_t^T R_{t+1} - \alpha)^+]\right).$$

Finally, we observe that $F_\beta(x_t, \alpha)$ is jointly convex in (x_t, α). The term α is linear (hence convex). The inner map $(x_t, \alpha) \mapsto (-x_t^\top R_{t+1} - \alpha)$ is affine. The outer function $z \mapsto z^+$ is convex and nondecreasing. The composition of a convex nondecreasing function with an affine map is convex. Thus $(-x_t^\top R_{t+1} - \alpha)^+$ is jointly convex in (x_t, α). Taking the expectation preserves convexity.

Since the objective function is convex, any local minimum is global, ensuring that optimization algorithms will converge efficiently to the optimal solution.

A.1 Walk-Forward Backtesting Framework

Building upon the continuous equivalence derived in Appendix A and following the optimization framework in the original paper (Ulyrch, Lucescu 2026), this section details the precise matrix formulations required to solve the regularized multi-currency Expected Shortfall models. By linearizing the L_1 penalty and discretizing the ES computations, we map both the separate overlay and joint optimization frameworks into standard Quadratic Programs of the form

$$\min_{\tilde{z}} \left(\tilde{c}^T \tilde{z} + \frac{1}{2} \tilde{z}^T \tilde{Q} \tilde{z} \right) \quad \text{subject to} \quad \tilde{D}\tilde{z} = \tilde{E}, \quad \tilde{A}\tilde{z} \leq b, \quad \tilde{z} \geq 0, \tag{A.1}$$

where $\tilde{z}$ is an augmented decision vector containing the portfolio weights and all necessary auxiliary variables.

A.2 Sample-Based Expected Shortfall

In practice, the conditional expectation from the ES objective is approximated using q historical joint return observations. Let $R_{\text{data}} \in \mathbb{R}^{q \times N}$ be the matrix of historical asset returns, where each row R_k^T represents the return vector at time k. The sample-based objective becomes:

$$\min_{x_t, \alpha} \left(-\mu^T x_t + \gamma\alpha + \frac{\gamma}{q(1-\beta)} \sum_{k=1}^{q} (-R_k^T x_t - \alpha)^+ \right).$$

To linearize the maximum operator $(\cdot)^+$, Rockafellar and Uryasev (2000) introduce a q-dimensional vector of auxiliary variables $u \in \mathbb{R}^q$, where $u_k = (-R_k^T x_t - \alpha)^+$. This implies two conditions for each sample k:

$$\begin{aligned} u_k &\geq 0 \\ u_k &\geq -R_k^T x_t - \alpha. \end{aligned}$$

Rearranging the second condition to isolate the variables yields a standard linear inequality:

$$-R_k^T x_t - \alpha - u_k \leq 0 \quad \forall\, k \in \{1, \ldots, q\}. \tag{A.2}$$

In block matrix notation across all q samples, this becomes:

$$-R_{\text{data}} x_t - \mathbf{1}_q \alpha - I_q u \leq \mathbf{0}_q,$$

where $\mathbf{1}_q$ is a q-dimensional vector of ones, I_q is the $q \times q$ identity matrix, and $\mathbf{0}_q$ is a q-dimensional vector of zeros. This constructs the first three block columns of the general inequality constraint matrix $\tilde{A}$.

A.3 Linearization of the L_1 Penalty and Leverage

The L_1 norm $\|x_t\|_1 = \sum_{i=1}^{N} |x_{i,t}|$ introduces non-differentiability into the objective function. To resolve this, we introduce non-negative auxiliary variables representing the positive and negative components of each asset weight: $\Delta_t^+, \Delta_t^- \in \mathbb{R}^N$ such that $\Delta_t^+ \geq 0$ and $\Delta_t^- \geq 0$.

The weights are decomposed as:

$$x_t = \Delta_t^+ - \Delta_t^- \implies I_N x_t + I_N \Delta_t^- - I_N \Delta_t^+ = \mathbf{0}_N. \tag{A.3}$$

Because the optimization minimizes the penalties, at most one of $\Delta_{i,t}^+$ or $\Delta_{i,t}^-$ will be non-zero for any asset i. Therefore, the absolute value is exactly determined by their sum: $|x_{i,t}| = \Delta_{i,t}^+ + \Delta_{i,t}^-$.

This allows the L_1 penalty term to be formulated linearly in the objective function as $\lambda_1 \mathbf{1}_N^T \Delta_t^+ + \lambda_1 \mathbf{1}_N^T \Delta_t^-$. Furthermore, the portfolio leverage constraint $\|x_t\|_1 \leq \ell$ transforms into a linear inequality:

$$\mathbf{1}_N^T \Delta_t^+ + \mathbf{1}_N^T \Delta_t^- \leq \ell. \tag{A.4}$$

A.4 Quadratic Translation of the L_2 Penalty

Unlike the expected shortfall and L_1 terms which map to the linear objective vector $\tilde{c}$, the L_2 penalty integrates naturally into the quadratic matrix $\tilde{Q}$. The penalty is defined as:

$$\lambda_2 \|x_t\|_2^2 = \lambda_2 x_t^T x_t = \frac{1}{2} x_t^T (2\lambda_2 I_N) x_t.$$

Thus, the block of $\tilde{Q}$ corresponding to the weights x_t is populated with the diagonal matrix $2\lambda_2 I_N$, while all auxiliary variables receive zeros.

A.5 Separate Optimization (Currency Overlay)

In the separate approach, asset weights $x_t \in \mathbb{R}^N$ and currency exposures $\psi_t \in \mathbb{R}^M$ are optimized sequentially.

A.5.1 Step 1: Asset-Level Optimization

The target vector relies on fully hedged asset returns $R^{\text{fh}} \in \mathbb{R}^{q \times N}$. We define the augmented vector of dimension $(3N + q + 1)$:

$$\tilde{x}_t = \left(x_t^T, \quad \alpha, \quad u^T, \quad (\Delta_t^-)^T, \quad (\Delta_t^+)^T \right)^T.$$

The objective vector $\tilde{c}$ and quadratic matrix $\tilde{Q}$ are constructed as:

$$\tilde{c} = \begin{pmatrix} -\mu_{\text{fh}} \\ \gamma \\ \frac{\gamma}{q(1-\beta)} \mathbf{1}_q \\ \lambda_1^a \mathbf{1}_N \\ \lambda_1^a \mathbf{1}_N \end{pmatrix}, \quad \tilde{Q} = \begin{pmatrix} 2\lambda_2^a I_N & O \\ O & O \end{pmatrix},$$

where O represents zero matrices of appropriate dimensions. The equality constraints enforce the budget condition $\mathbf{1}_N^T x_t = 1$ and the variable decomposition from Equation A.3:

$$\tilde{D} = \begin{pmatrix} \mathbf{1}_N^T & 0 & \mathbf{0}_q^T & \mathbf{0}_N^T & \mathbf{0}_N^T \\ I_N & \mathbf{0}_N & O_{N \times q} & I_N & -I_N \end{pmatrix}, \quad \tilde{E} = \begin{pmatrix} 1 \\ \mathbf{0}_N \end{pmatrix}.$$

The inequality matrix $\tilde{A}$ and bound vector b encode the shortfall losses from Equation A.2 and the leverage limit from Equation A.4:

$$\tilde{A} = \begin{pmatrix} -R^{\text{fh}} & -\mathbf{1}_q & -I_q & O_{q \times N} & O_{q \times N} \\ \mathbf{0}_N^T & 0 & \mathbf{0}_q^T & \mathbf{1}_N^T & \mathbf{1}_N^T \end{pmatrix}, \quad b = \begin{pmatrix} \mathbf{0}_q \\ \ell \end{pmatrix}.$$

A.5.2 Step 2: Currency-Level Optimization

Conditional on the optimal asset weights x_t^*, the currency exposures ψ_t are optimized. Using historical excess currency returns $R^c \in \mathbb{R}^{q\times M}$, the augmented vector becomes $\tilde{\psi}_t \in \mathbb{R}^{(3M+q+1)}$.

Because currency overlays act as zero-investment portfolios, there is no budget constraint. Additionally, the pre-fixed asset portfolio returns ($R^{\text{fh}}x_t^*$) must be subtracted from the shortfall inequality constraint boundaries. Thus, the formulations shift to:

$$\tilde{D}_\psi = \begin{pmatrix} I_M & \mathbf{0}_M & O_{M\times q} & I_M & -I_M \end{pmatrix}, \quad \tilde{E}_\psi = \mathbf{0}_M,$$

$$\tilde{A}_\psi = \begin{pmatrix} -R^c & -\mathbf{1}_q & -I_q & O_{q\times M} & O_{q\times M} \end{pmatrix}, \quad b_\psi = R^{\text{fh}}x_t^*.$$

A.6 Joint Optimization Framework

The joint formulation merges assets and currencies into a single decision vector $\theta_t = (x_t^T, \psi_t^T)^T \in \mathbb{R}^{N+M}$, evaluated over the joint return block matrix $R^{\text{all}} = \begin{pmatrix} R^{\text{fh}} & R^c \end{pmatrix} \in \mathbb{R}^{q\times(N+M)}$.

We construct an expansive augmented vector $\tilde{\theta}_t \in \mathbb{R}^{(3N+3M+q+1)}$:

$$\tilde{\theta}_t = \begin{pmatrix} x_t^T, & \psi_t^T, & \alpha, & u^T, & (\Delta_{x,t}^-)^T, & (\Delta_{\psi,t}^-)^T, & (\Delta_{x,t}^+)^T, & (\Delta_{\psi,t}^+)^T \end{pmatrix}^T.$$

The objective vector applies separate L_1 penalty parameters (λ_1^a for assets, λ_1^c for currencies):

$$\tilde{c} = \begin{pmatrix} -\mu_{\text{fh}} \\ -\mu_c \\ \gamma \\ \frac{\gamma}{q(1-\beta)}\mathbf{1}_q \\ \lambda_1^a\mathbf{1}_N \\ \lambda_1^c\mathbf{1}_M \\ \lambda_1^a\mathbf{1}_N \\ \lambda_1^c\mathbf{1}_M \end{pmatrix}, \quad \tilde{Q} = \begin{pmatrix} 2\lambda_2^a I_N & O & O \\ O & 2\lambda_2^c I_M & O \\ O & O & O \end{pmatrix}.$$

The budget constraint applies strictly to the asset weights, while the decomposition constraints apply to both assets and currencies.

$$\tilde{D} = \begin{pmatrix} \mathbf{1}_N^T & \mathbf{0}_M^T & 0 & \mathbf{0}_q^T & \mathbf{0}_{N+M}^T & \mathbf{0}_{N+M}^T \\ I_{N+M} & & \mathbf{0}_{N+M} & O_{(N+M)\times q} & I_{N+M} & -I_{N+M} \end{pmatrix}, \quad \tilde{E} = \begin{pmatrix} 1 \\ \mathbf{0}_{N+M} \end{pmatrix}.$$

Finally, the inequality matrix binds the unified shortfall distribution and enforces the leverage limit solely on the gross asset exposure:

$$\tilde{A} = \begin{pmatrix} -R^{\text{all}} & -\mathbf{1}_q & -I_q & O_{q\times(2N+2M)} & & \\ \mathbf{0}_{N+M}^T & 0 & \mathbf{0}_q^T & \begin{pmatrix}\mathbf{1}_N^T & \mathbf{0}_M^T\end{pmatrix} & \begin{pmatrix}\mathbf{1}_N^T & \mathbf{0}_M^T\end{pmatrix} & \end{pmatrix}, \quad b = \begin{pmatrix} \mathbf{0}_q \\ \ell \end{pmatrix}.$$

Solving this comprehensive quadratic program yields the optimal regularized asset and currency allocations concurrently.

Appendix B

Optimization Derivation: Mean-Variance

Appendix C

Transaction Costs

Following the methodology outlined by [4], [4] and [**?**], and keeping the notation introduced in Chapter **??**, we model transaction costs as proportional to the volume of traded assets and currencies, ignoring any fixed trading costs. The return of the hedged portfolio is expressed as the sum of unhedged asset returns and gains from forward currency positions

$$R^h_{P,t+1} = \sum_{i=1}^{N} x_{i,t} R^u_{i,t+1} + \sum_{c=1}^{M} \varphi_{c,t}(f_{c,t} - e_{c,t+1}). \tag{(3.2)}$$

Let τ^{asset}_{t+1} and τ^{curr}_{t+1} denote the total transaction costs associated with asset and currency trades, respectively. The portfolio return net of transaction costs is then given by

$$R^{h,net}_{P,t+1} = R^h_{P,t+1} - \tau^{asset}_{t+1} - \tau^{curr}_{t+1}. \tag{C.1}$$

The total transaction costs are computed as the sum of the costs incurred from trading individual assets and currencies, weighted by the respective trading volumes

$$\tau^{asset}_{t+1} = \sum_{i=1}^{N} \tau^{asset}_{i,t+1} |x_{i,t+1} - x^-_{i,t+1}|,$$

$$\tau^{curr}_{t+1} = \sum_{c=1}^{M} \tau^{curr}_{c,t+1} \varphi_{c,t+1} + \sum_{c=1}^{M} \tau^{curr}_{c,t+1} |w_{c,t+1} - w^-_{c,t+1}|,$$

where $\tau^{asset}_{i,t+1}$ and $\tau^{curr}_{c,t+1}$ are the per-unit transaction costs for asset i and currency c. The terms $x^-_{i,t+1}$ and $w^-_{c,t+1}$ denote the asset weights and implied currency weights immediately prior to rebalancing at time $t+1$ (i.e., the drifted weights).

Asset transaction costs are incurred exclusively when rebalancing individual asset holdings. Currency transaction costs, however, arise from two sources, adjustments in forward hedge positions (assuming the forward contract maturity matches the rebalancing frequency, the old contract expires and a new contract of full notional size $\varphi_{c,t+1}$ must be established at each rebalancing date) and currency exchanges required to fund purchases of assets denominated in different currencies ($|w_{c,t+1} - w^-_{c,t+1}|$). If the portfolio is rebalanced more frequently than the forward contract maturity, existing forward contracts are kept open and only incremental adjustments are traded. In this case, the forward transaction cost term is modeled as the absolute difference, replacing $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} \varphi_{c,t+1}$ with $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} |\varphi_{c,t+1} - \varphi^-_{c,t+1}|$, where $\varphi^-_{c,t+1}$ represents the drifted hedge weight. Following [4], these costs are aggregated at the currency level, operating under the assumption that proceeds from selling assets denominated in

currency c can fund the purchase of other assets in the same currency without incurring currency conversion fees.

To define the individual per-unit costs, we consider an asset i denominated in currency c_i. Let $P^{ask}_{i,t}$, $P^{bid}_{i,t}$, and $P_{i,t}$ represent the ask, bid, and mid prices of the asset, while $S^{ask}_{c_i,t}$, $S^{bid}_{c_i,t}$, and $S_{c_i,t}$ denote the corresponding currency prices. If asset i is purchased at time t and sold at time $t+1$, its realized log return is given by

$$\begin{aligned}
\log\left(\frac{P^{bid}_{i,t+1}S^{bid}_{c_i,t+1}}{P^{ask}_{i,t}S^{ask}_{c_i,t}}\right) &= \log\left(\frac{\left(P_{i,t+1}-\frac{\delta^{asset}_{i,t+1}}{2}\right)\left(S_{c_i,t+1}-\frac{\delta^{curr}_{c_i,t+1}}{2}\right)}{\left(P_{i,t}+\frac{\delta^{asset}_{i,t}}{2}\right)\left(S_{c_i,t}+\frac{\delta^{curr}_{c_i,t}}{2}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}\left(1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}\right)S_{c_i,t+1}\left(1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}\right)}{P_{i,t}\left(1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}\right)S_{c_i,t}\left(1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)+\log\left(\frac{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}\right)+\log\left(\frac{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)-\log\left(\frac{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}\right)-\log\left(\frac{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}\right),
\end{aligned}$$

where δ^{asset}_i and $\delta^{curr}_{c_i}$ are the bid-ask spreads. Comparing this to equation (C.1), the transaction cost is precisely the difference between the actual realized log return and mid-price returs for the asset and currency.

Defining half of the proportional bid-ask spreads as

$$\chi^{asset}_{i,t}=\frac{P^{ask}_{i,t}-P^{bid}_{i,t}}{2P_{i,t}} \quad \text{and} \quad \chi^{curr}_{c,t}=\frac{S^{ask}_{c,t}-S^{bid}_{c,t}}{2S_{c,t}},$$

the the per-unit transaction costs for trading asset i and currency c become

$$\tau^{asset}_{i,t+1}=\begin{cases}\log\left(\frac{1+\chi^{asset}_{i,t+1}}{1-\chi^{asset}_{i,t}}\right) & \text{if } x_{i,t+1}>x^{-}_{i,t+1},\\ \log\left(\frac{1-\chi^{asset}_{i,t}}{1+\chi^{asset}_{i,t+1}}\right) & \text{otherwise,}\end{cases}$$

and

$$\tau^{curr}_{c,t+1}=\begin{cases}\log\left(\frac{1+\chi^{curr}_{c,t+1}}{1-\chi^{curr}_{c,t}}\right) & \text{if } \varphi_{c,t+1}>0 \text{ or } w_{c,t+1}>w^{-}_{c,t+1},\\ \log\left(\frac{1-\chi^{curr}_{c,t}}{1+\chi^{curr}_{c,t+1}}\right) & \text{otherwise.}\end{cases}$$

For our empirical analysis, we adopt the simplifying assumption, that proportional bid-ask spreads remain constant both over time and across the cross-sections of both assets and currencies. We set the half-spreads to $\chi^{asset}=0.001$ (10 bps) for all equities and $\chi^{curr}=0.0001$ (1 bp) for all currencies, consistent with empirical finidngs in [**?**] and [**?**]. Under these assumptions, the transaction costs simplify to constant values of $\tau^{asset}\approx 0.002$ and $\tau^{curr}\approx 0.0002$.

Appendix D

Hyperparameter Tuning and Additional Results

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